

FSK116: Tutorial Problem Numbers / Tutoriaal Probleem Nommers

2017

6 Week: 27 March 2017 / 27 Maart 2017

All the indicated numbers refer to the “Problems” section in Serway; e.g. P1.9 is problem 9 from chapter 1.

Alle nommers verwys na die “Problems” afdeling in Serway; bv. P1.9 is probleem 9 uit hoofstuk 1.

P7.9 16.0

P7.11 (a) 16.0 J (b) 36.9°

P7.13 7.05 m at 28.4°

P7.14 (a) 24.0 J (b) -3.00 J (c) 21.0 J

P7.15 (a) 7.50 J (b) 15.0 J (c) 7.50 J (d) 30.0 J

P7.19 (a) 575 N/m (b) 46.0 J

P7.26 (a) Graph (b) -12.0 J

P7.27 (a) Graph (b) approximately 116 N/m (c) Explanation (d) Slope of your graph (e) With $k = 116$ N/m, the force is 12.2 N

P7.28 (a) 9.00 kJ (b) 11.7 kJ (c) Work is greater by 29.6%

P7.58 $W = k_1 \frac{x_{\max}^2}{2} + k_2 \frac{x_{\max}^3}{3}$